

iPhone 11 Disassembly — Interactive HTML Artifact Build Plan

Status: Ready to build. Waiting on user-provided images of the interior (logic board side) and backplate before implementation begins.

1. Project Overview

Goal: A single self-contained `.html` artifact that simulates an iPhone 11 teardown. The user clicks parts to remove them one at a time in correct disassembly order. Each removed part triggers a detailed info card below the diagram. A persistent back button undoes removals one step at a time (full undo history). The diagram visually references the user's actual physical device via embedded photos.

Delivery method: Cowork artifact (persistent, re-openable HTML page)

2. Image Integration Instructions

When the user provides images, use them as follows:

Image provided	How to use it
Backplate photo (exterior rear of phone)	Use as a texture/background reference for the chassis SVG layer at the bottom of the exploded stack
Interior / logic board side photo	Crop and embed as a <code><image></code> tag inside the SVG for the logic board layer — parts are then drawn as highlight overlays on top of it
Display assembly interior photo	Embed as the display layer SVG background
Any other disassembly step photos	Use as layer backgrounds for the corresponding exploded stack level

Implementation approach for photos:

- Convert each image to a Base64 data URI and embed directly in the HTML (no external file dependencies)
- Use `<image>` tags within SVG layers, with parts drawn as semi-transparent colored SVG overlays on top
- Keep SVG overlay opacity at ~0.4 so the real photo shows through but the interactive part boundaries are still visible
- When a part is "removed" (clicked), fade its SVG overlay out — the underlying photo layer stays visible so the user can see what's beneath

3. Visual Design Specifications

Layout

iPhone 11 Teardown	[← Back] [12/43 removed]
[Exploded Diagram — center stage]	
Parts stack vertically, separated	
by ~20px gaps, phone outline ghost	
behind all layers	
[Info card — slides in below on click]	
Part Name Category badge	

Function Connections Specs Fun facts	

Color coding by category

Category	Color
Display / Glass	#5BC8F5 (sky blue)
Cameras	#A78BFA (purple)
Logic / Chips	#34D399 (green)
Power / Battery	#FCD34D (amber)
Audio	#F87171 (red)
Connectivity / Antenna	#60A5FA (blue)
Mechanical / Screws / Brackets	#94A3B8 (slate)
Sensors	#FB923C (orange)
Flex cables	#E879F9 (pink)

Exploded stack order (top = first removed, bottom = chassis)

Layers are spaced ~24px apart vertically in the SVG. The phone ghost outline (thin gray stroke, no fill) sits behind all layers as a positioning reference.

4. Complete Parts List (43 Parts)

Parts are listed in disassembly order. Each entry includes the info card content to display when clicked.

LAYER 0 — External Fasteners

1. Pentalobe Screw × 2 (bottom edge)

- **Function:** Sole external fasteners holding the display assembly to the chassis. Apple uses a proprietary 5-pointed Pentalobe head to deter DIY repair.
- **Connects to:** Display assembly frame, bottom of chassis
- **Specs:** 1.2mm Pentalobe, ~3.9mm length
- **Tool required:** P2 Pentalobe screwdriver

- **Fun fact:** Apple introduced Pentalobe screws in 2011 specifically to prevent user-serviceable repairs. No standard screwdriver fits them.
-

LAYER 1 — Display Assembly

2. Display Assembly (full unit)

- **Function:** Everything you see and touch — houses the LCD panel, digitizer (touch layer), front-facing sensors, and earpiece. On iPhone 11, this is an IPS LCD (not OLED).
- **Connects to:** Logic board (display connector, touch connector, front sensor flex), chassis frame
- **Specs:** 6.1" IPS LCD, 1792×828 resolution, 326 ppi, 625 nits brightness, True Tone
- **Fun fact:** iPhone 11 was the last new iPhone model to ship with LCD instead of OLED. Apple called it "Liquid Retina."

3. LCD Panel

- **Function:** Produces the actual image using a backlight + liquid crystal matrix. Unlike OLED, it requires a separate backlight layer.
- **Connects to:** Digitizer layer (bonded), display flex cable to logic board
- **Specs:** IPS LCD, 1792×828, 60Hz refresh

4. Digitizer (Touch Layer)

- **Function:** Capacitive touch sensor bonded to the front of the LCD. Detects finger position by measuring changes in an electrical field grid.
- **Connects to:** LCD (bonded), touch flex cable to logic board
- **Note:** On iPhone 11, digitizer and LCD are not fused to the outer glass — they are fused to each other but the outer glass is separate, making independent glass replacement difficult.

5. Earpiece Speaker Mesh

- **Function:** Acoustically transparent mesh covering the earpiece speaker slot. Keeps debris out while allowing sound through. Also used as a secondary speaker in stereo mode.
- **Connects to:** Display assembly frame (press fit)

6. Ambient Light / Proximity Sensor Flex

- **Function:** Two sensors on one flex cable. Proximity sensor (IR emitter + receiver) detects when the phone is held to your ear and turns off the display. Ambient light sensor adjusts screen brightness automatically.
- **Connects to:** Front sensor module, logic board via sensor flex

LAYER 2 — Face ID / Front Camera Module

7. TrueDepth Camera Assembly (full bracket)

- **Function:** The entire notch module. Contains all Face ID and front camera components on a single bracket. Removing this one bracket takes out 7 functional components at once.
- **Connects to:** Logic board via front sensor cable, display assembly housing

8. Front Camera (12MP)

- **Function:** Captures selfie photos and video. Also enables Portrait mode using software depth-mapping (no dedicated depth sensor on the front).
- **Specs:** 12MP, f/2.2 aperture, 23mm equiv., 4K video @ 60fps, 120fps slo-mo @ 1080p
- **Connects to:** TrueDepth flex cable → logic board

9. Dot Projector

- **Function:** Projects 30,000+ invisible infrared dots onto your face to create a 3D depth map. Core of the Face ID system.
- **Connects to:** Secure Enclave on A13 Bionic via encrypted bus
- **Fun fact:** The dot pattern is unique to each projector and calibrated at the factory. Replacing just the dot projector without Apple's calibration software will break Face ID permanently.

10. Flood Illuminator

- **Function:** Emits a burst of infrared light to illuminate your face in the dark so the IR camera can see you even with no visible light.
- **Connects to:** TrueDepth flex, logic board

11. Infrared Camera

- **Function:** Reads the dot pattern from the dot projector and the flood-illuminated IR image of your face. Sends both to the Secure Enclave for comparison against stored Face ID data.
- **Connects to:** A13 Bionic Secure Enclave (encrypted)

12. Microphone (front/top)

- **Function:** Primary microphone for calls and voice input when phone is held upright. Also used for noise cancellation pairing with the bottom mic.
 - **Connects to:** Audio codec on logic board
-

LAYER 3 — Battery Connector Area

13. Battery Connector Bracket

- **Function:** Metal plate that screws down over the battery connector to prevent it from vibrating loose. Must be removed before disconnecting the battery.
- **Connects to:** Logic board, chassis
- **Screws:** 2× Phillips #000

14. Battery Connector (flex end)

- **Function:** The ZIF (zero insertion force) connector that carries power from the battery to the logic board. Always disconnect this first before touching anything else inside — it prevents short circuits.
 - **Connects to:** Battery → logic board PMIC
-

LAYER 4 — Battery

15. Battery

- **Function:** Lithium-ion rechargeable cell. Powers everything. iPhone 11 uses a single L-shaped cell.
 - **Specs:** 3110 mAh, 3.83V nominal, 11.91Wh. Supports 18W fast charge (charger sold separately) and Qi wireless charging (7.5W with Apple, 5W generic).
 - **Connects to:** Logic board via battery connector flex
 - **Adhesive:** Secured with two pull-tab adhesive strips (similar to AirPower concept — pull tabs sideways to release)
 - **Fun fact:** Apple introduced "performance management" (throttling) for aged batteries after the 2017 controversy. Battery health is visible in Settings → Battery → Battery Health.
-

LAYER 5 — Taptic Engine

16. Taptic Engine

- **Function:** Linear resonant actuator (LRA) that produces precise haptic feedback. Replaces the old rotary vibration motor. Enables "clicks" on the solid-state Home button (iPhone 8) and subtle UI feedback throughout iOS.
- **Specs:** Custom Apple LRA, moves a weighted mass back and forth on a spring at precise frequencies
- **Connects to:** Logic board via haptic flex cable

- **Fun fact:** The Taptic Engine is so precisely tuned that it can simulate the feeling of clicking a physical button that doesn't actually move (used on the Home button on iPhone 7+).
-

LAYER 6 — Audio & Charging

17. Bottom Speaker Assembly

- **Function:** Main media speaker (louder of the two stereo speakers). Works in tandem with the earpiece speaker for stereo audio. Housed in a sealed enclosure within the chassis.
- **Specs:** Stereo pairing with earpiece, up to 3dB louder than iPhone XR
- **Connects to:** Logic board via speaker flex

18. Charging Port Flex Cable

- **Function:** Combines multiple components: Lightning connector, bottom microphone, antenna connection points, SIM card tray slot, and the secondary speaker grille. One of the most complex single flex cables in the phone.
- **Connects to:** Logic board (multi-point connection), battery (charging path), bottom speaker, antennas
- **Fun fact:** Apple uses Lightning on iPhone 11. The EU forced a switch to USB-C starting with iPhone 15 (2023).

19. Bottom Microphone

- **Function:** Secondary microphone on the charging flex. Used during speakerphone calls, video recording, and noise cancellation. Paired with top mic for directional audio.
 - **Connects to:** Audio codec via charging port flex
-

LAYER 7 — Logic Board

20. Logic Board (full assembly)

- **Function:** The brain of the phone. A small PCB (printed circuit board) densely packed with chips, using Apple's proprietary SiP (System in Package) stacking to fit multiple chips in a tiny footprint.
- **Connects to:** Everything — display, battery, cameras, speakers, sensors, antennas
- **Size note:** iPhone 11 uses a two-layer stacked logic board design introduced with iPhone X, making it about 70% the area of a conventional single-layer board.

21. Apple A13 Bionic (SoC)

- **Function:** System on Chip — contains the CPU, GPU, Neural Engine, Secure Enclave, ISP (image signal processor), and memory controller all on a single die.
- **Specs:** 7nm TSMC process, 6-core CPU (2 performance + 4 efficiency), 4-core GPU, 16-core Neural Engine (600 billion operations/sec), 4GB LPDDR4X RAM
- **Connects to:** All other chips on logic board, sensors, cameras
- **Fun fact:** The A13's Neural Engine is what enables real-time computational photography features like Deep Fusion, Smart HDR, and night mode — running up to 1 trillion operations per photo.

22. NAND Flash Storage

- **Function:** Non-volatile storage for your apps, photos, OS, and data. Uses 3D NAND technology — cells are stacked vertically to pack more storage in less space.
- **Specs:** 64GB / 128GB / 256GB depending on model. Manufacturer: Toshiba or Samsung (varies by production run)
- **Connects to:** A13 Bionic via NVMe interface

23. LPDDR4X RAM (stacked with A13)

- **Function:** Working memory — holds the OS, open apps, and data being actively processed. iPhone 11 has 4GB. When RAM fills up, iOS has to reload apps from NAND (the "app reload" behavior).
- **Specs:** 4GB LPDDR4X, stacked directly on top of the A13 die in a Package on Package (PoP) configuration
- **Connects to:** A13 Bionic (direct die-to-die connection)

24. Power Management IC (PMIC)

- **Function:** Regulates all the different voltages needed by different components. Converts battery voltage (~3.8V) into dozens of precise voltages (0.8V for CPU, 1.8V for memory, etc.). Also manages charging via USB and wireless.
- **Manufacturer:** Apple custom / Dialog Semiconductor (varies)
- **Connects to:** Battery, A13, every major component

25. Intel / Qualcomm Modem

- **Function:** Handles all cellular communication — LTE data, voice calls, SMS. iPhone 11 uses an Intel XMM7660 modem (non-5G). This was the last iPhone generation to use Intel modems; Apple switched to Qualcomm with iPhone 12.
- **Specs:** Cat.18 LTE, up to 1.6 Gbps download
- **Fun fact:** Apple sued Qualcomm in 2017 over licensing fees, switched to Intel modems for two years, found Intel's modems inferior in benchmarks, then settled with Qualcomm and switched back for iPhone 12.

26. Wi-Fi / Bluetooth / Ultra Wideband Chip

- **Function:** Three radios on one chip. Wi-Fi (802.11ax / Wi-Fi 6 not on iPhone 11 — it uses 802.11ac Wi-Fi 5), Bluetooth 5.0, and Ultra Wideband (U1 chip for spatial awareness / AirDrop precision).
- **Manufacturer:** Apple U1 for UWB; Broadcom for Wi-Fi/BT
- **Connects to:** Antennas, A13, charging flex (for antenna switching)

27. NFC Controller

- **Function:** Near Field Communication — enables Apple Pay, transit cards, reading NFC tags, and background tag reading. Range: ~5cm.
- **Manufacturer:** NXP Semiconductors
- **Connects to:** Antennas, Secure Enclave (for payment security)

28. Audio Codec

- **Function:** Converts digital audio signals from the A13 into analog signals for the speakers, and converts analog signals from the microphones into digital data. Acts as the phone's dedicated sound card.
- **Manufacturer:** Cirrus Logic CS42L83
- **Connects to:** A13, all microphones, speakers, Lightning port (for wired audio)

29. Barometric Vent / Pressure Sensor

- **Function:** Tiny MEMS sensor that measures atmospheric pressure. Used for elevation data in Maps, fitness tracking (floor counting), and weather apps. Also helps maintain IP68 seal integrity data.
 - **Manufacturer:** Bosch / STMicroelectronics
 - **Connects to:** A13 via I2C bus
-

LAYER 8 — Rear Camera System

30. Rear Camera Module (full bracket)

- **Function:** Dual-camera bracket assembly holding both rear cameras, the True Tone flash, and the rear microphone in a single removable unit.
- **Connects to:** Logic board via camera flex cables (separate connectors for each camera)

31. Wide Camera (12MP, main)

- **Function:** Primary camera. Used for most photos and videos. Has optical image stabilization (OIS) — a gyroscope physically tilts the lens to counteract hand shake.
- **Specs:** 12MP, f/1.8 aperture, 26mm equiv., OIS, 4K60 video
- **Connects to:** ISP in A13 Bionic

32. Ultra Wide Camera (12MP)

- **Function:** 0.5× zoom camera. Captures a 120-degree field of view for landscapes, architecture, and group shots. No OIS (fixed lens).
- **Specs:** 12MP, f/2.4 aperture, 13mm equiv., 120° FOV
- **Connects to:** ISP in A13 Bionic
- **Fun fact:** The ultra-wide camera was introduced on iPhone 11 — the first iPhone to have it. It's the same lens used in the "slofie" (slow motion selfie) feature.

33. True Tone Flash (Quad-LED)

- **Function:** Four LEDs in two pairs — two "warm" and two "cool" color temperature LEDs. The A13's ISP analyzes the scene and adjusts the ratio of warm/cool LEDs to match the ambient light color, producing more natural-looking flash photos.
- **Connects to:** Logic board, PMIC (for power surge management during flash)

34. Rear Microphone

- **Function:** Third microphone on the device. Positioned near the rear cameras to capture audio during video recording with minimal wind noise and maximum directional accuracy.
 - **Connects to:** Audio codec via camera flex
-

LAYER 9 — Connectivity

35. SIM Card Tray

- **Function:** Houses the nano-SIM card. Also contains the SIM card reader contacts that route to the logic board. iPhone 11 supports a physical SIM + 1 eSIM (not dual physical SIM in US models).
- **Connects to:** Logic board via SIM flex traces in chassis
- **Note:** Ejected with a SIM tool (1mm pin) via the side slot — does not require opening the phone.

36. Upper Antenna Board

- **Function:** One of several antenna modules. Handles specific LTE bands and Wi-Fi signals routed to the modem and Wi-Fi chip. Apple uses antenna switching to dynamically route signals to the best-positioned antenna based on how you're holding the phone.
- **Connects to:** Modem, Wi-Fi chip via coaxial RF cables

37. Lower Antenna Board

- **Function:** Handles additional LTE bands and the NFC antenna loop. Positioned at the bottom of the chassis near the Lightning port.

- **Connects to:** NFC controller, modem

38. Rear-Facing Antenna (embedded in glass back)

- **Function:** Enables wireless charging reception. Also handles Bluetooth and some Wi-Fi bands. Printed directly onto the interior of the glass back panel.
 - **Connects to:** PMIC (for charging), Wi-Fi/BT chip
-

LAYER 10 — Chassis / Midframe

39. Volume Up Button Flex

- **Function:** Physical button + flex cable for the upper volume button on the left side. Sends a digital signal to the A13 on press.
- **Connects to:** Logic board via side button flex

40. Volume Down Button Flex

- **Function:** Same as above for the lower volume button.
- **Connects to:** Logic board via side button flex

41. Mute / Ring Switch Flex

- **Function:** The physical toggle switch above the volume buttons. Interrupts the ringer and notification audio path at a hardware level — not dependent on software.
- **Connects to:** Logic board

42. Side Button (Power / Siri / Apple Pay) Flex

- **Function:** The right-side button. Handles: wake/sleep, power off, emergency SOS (hold + volume), Face ID payment confirmation, and Siri activation.
- **Connects to:** Logic board

43. Chassis / Midframe

- **Function:** The structural backbone of the entire phone. Made from 7000-series aluminum (aerospace grade). Every other component mounts to or against it. Provides structural rigidity, heat dissipation, and the antenna break lines (black/white lines that interrupt the metal frame to let radio signals pass through).
 - **Material:** 7075 aluminum alloy
 - **Fun fact:** The antenna break lines (the thin non-metal lines in the aluminum frame) are actually structural compromises Apple engineers work around — too many breaks weaken the frame, too few block the radio signals.
-

5. Interaction Design (Detailed)

Disassembly

- Parts are rendered as SVG shapes (filled regions + labels) on top of the photo layer
- Only the **currently removable** part(s) should be highlighted/pulsing — parts below in the stack are locked until upper parts are removed (enforces correct disassembly order)
- Hovering a removable part: show a tooltip with part name + category badge
- Clicking a removable part:
 1. Play a brief CSS animation — part slides up and fades out
 2. Part moves to a "removed parts tray" (small thumbnail strip on the side or below)
 3. Info card slides in below the diagram with full part details
 4. Part counter updates: "12 / 43 removed"

Undo (Back Button)

- Always-visible button top-left:  Back
- Each click restores the last removed part (reverse animation — slides back in from top)
- Full stack-based undo history — can undo all the way back to the start
- Disabled (grayed out) if nothing has been removed yet

Info Card

- Slides in below the diagram on click (CSS transition)
- Contains:
 - Part name (large)
 - Category badge (color-coded)
 - **Function** section
 - **Connects to** section (listed components)
 - **Specs** section (where applicable)
 - **Fun fact** section (highlighted in a callout box)
- Dismiss button (×) or clicking another part replaces it

Completion State

- When all 43 parts are removed, show a congratulations screen: "Full teardown complete! You've disassembled an iPhone 11 into all [43] components."
 - Option to "Reassemble" (play the whole undo history back in reverse, animated)
-

6. Technical Implementation

File structure

Single self-contained `.html` file — no external dependencies except optionally loading from CDN:

- **No CDN needed** — all JS and CSS are inline

SVG approach

```
<svg id="diagram" viewBox="0 0 375 812">
```

```
<!-- Ghost phone outline -->
```

```
<rect ... class="phone-ghost" />
```

```
<!-- Each layer is a <g> group with a data-part-id attribute -->
```

```
<g class="part-layer" data-part-id="2" data-layer="1" data-unlocked="true">
```

```
<!-- Background: <image> tag with base64 photo -->
```

```
<image href="data:image/jpeg;base64,..." ... opacity="0.9" />
```

```
<!-- Interactive overlay shape -->
```

```
<path class="part-overlay" d="..." fill="#5BC8F5" opacity="0.4" />
```

```
<text class="part-label">Display Assembly</text>
```

```
</g>
```

```
<!-- Repeat for each of 43 parts -->
```

```
</svg>
```

JavaScript state

```
const state = {
```

```
history: [],      // stack of removed part IDs (for undo)

removedParts: new Set(), // currently removed

selectedPart: null,  // currently showing info card

parts: PARTS_DATA    // array of 43 part objects (see below)

};
```

Part data object shape

```
{

  id: 15,

  name: "Battery",

  layer: 4,

  category: "power",

  color: "#FCD34D",

  svgPath: "M...",    // SVG path for the overlay shape

  photoLayer: "battery", // which base64 photo to use as background

  unlockedBy: [14],    // part IDs that must be removed first

  info: {

    function: "...",

    connectsTo: ["Logic board via battery connector flex"],

    specs: ["3110 mAh", "3.83V", "11.91Wh"],

    funFact: "..."

  }

}
```

Dependency graph (unlock order)

Pentalobe Screws (1)

↳ Display Assembly (2)

↳ LCD Panel (3), Digitizer (4), Earpiece Mesh (5), Ambient Sensor Flex (6)

↳ TrueDepth Assembly (7)

↳ Front Camera (8), Dot Projector (9), Flood Illuminator (10), IR Camera (11),
Front Mic (12)

↳ Battery Connector Bracket (13) → Battery Connector (14)

↳ Battery (15)

↳ Taptic Engine (16)

↳ Bottom Speaker (17) → Charging Port Flex (18) → Bottom Mic (19)

↳ Logic Board (20)

↳ [All chips 21-29 unlock simultaneously]

↳ Rear Camera Module (30)

↳ Wide Camera (31), Ultra Wide (32), Flash (33), Rear
Mic (34)

↳ SIM Tray (35), Upper Antenna (36), Lower Antenna
(37), Rear Antenna (38)

↳ Button flexes (39-42)

↳ Chassis (43)

7. When Images Are Provided

Do the following steps in order:

1. **Receive photos** from the user (front interior, rear interior/logic board, backplate)
2. **Identify physical features** visible in each photo:

- Logic board photo: locate the A13, cameras, battery connector, flex cables
 - Note the exact color of the user's phone (Black, White, Yellow, Purple, Green, Red)
 - Note any visible wear, modifications, or unique markings
3. **Convert each image to base64** (use Python in the workspace shell):

```
python3 -c "import base64;  
print(base64.b64encode(open('photo.jpg','rb').read()).decode())"
```

4. **Assign each photo** to the correct SVG layer as a `<image href="data:image/jpeg;base64,...">` tag
5. **Draw SVG overlay paths** that match the actual positions of parts visible in the photos
6. **Adjust part colors/labels** to match real positions in the user's specific unit
-

8. Build Checklist (for implementation session)

Receive and process user's photos (base64 encode)
Set up HTML skeleton with CSS variables and layout
Build SVG diagram with all 43 part layers
Embed photos as SVG image layers under overlays
Implement JS state machine (removed set, history stack)
Implement unlock/lock dependency logic
Add hover tooltip behavior
Add click → info card behavior
Add Back button (undo) behavior
Add part counter
Add removed parts tray
Add completion / reassemble state
Test full disassembly flow start to finish
Test full undo flow back to start
Verify all 43 info cards have content
Deliver as Cowork artifact

Plan version 1.0 — created May 14, 2026 Ready for implementation once user provides phone interior images.